



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ASTROPHYSICS				
Type of Course	Major Elective				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY4CJ205 Modern Physics				
Course Summary	This course gives a pedagogical introduction to astronomy and astrophysics by introducing the students the techniques to measure astronomical parameters, the properties of the Sun, stellar evolution and properties of galaxies and an overview of the Universe.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a deep understanding of theoretical frameworks in astronomy,	U	C	Instructor-created exams / Quiz

	including celestial mechanics, stellar structure, and cosmology.			
CO2	Apply basic physical principles from a broad range of topics in physics to address complex astronomical phenomena.	Ap	P	Viva Voce / Home Assignments/ Seminar Presentations
CO3	Get knowledge of positional astronomy, astronomical parameters and tools.	U	C	Instructor-created exams / Quiz
CO4	Able to explain the physics of Sun and the evolution of stars.	U	C	Instructor-created exams / Quiz
CO5	Describe the morphology and classification of galaxies and galaxy clusters.	U	C	Instructor-created exams / Quiz
CO6	Expose scientific knowledge about the origin and evolution of the universe.	U	C	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	ASTRONOMICAL PARAMETERS AND TOOLS		12	18
	1	The celestial sphere, The constellations, The celestial coordinate system	2	

	2	Stellar luminosity, Stellar distances, The parsec, The Cepheid variable distance scale	3	
	3	Stellar magnitudes, Apparent magnitudes, Magnitude calculations, The absolute magnitude scale, The standard formula to derive absolute magnitudes	3	
	4	Colour and surface temperature, Stellar photometry, Stellar spectra, Spectral types, Spectroscopic parallax.	2	
	5	Basics of refracting telescopes- Resolution, Magnification, Newtonian telescope	1	
	6	Active and adaptive optics, Active optics, Adaptive optics	1	
	Sections 1.3, 1.3.1, 1.3.2, 1.3.3, 1.3.4, 1.4, 5.3, 5.4, 5.5, 5.7, 5.10, 5.10.1, 5.10.2, 6.1, 6.2, 6.2.1, 6.4, 6.4.1, 6.5, 6.6, 6.7, 6.7.2, 6.8, 8.3.3 of Book 1			
II	THE SUN AND HR DIAGRAM		12	18
	7	The Sun, Overall properties of the Sun, The Sun's total energy output, The Fraunhofer lines in the solar spectrum	2	
	8	Nuclear fusion, The proton–proton cycle	2	
	9	The solar neutrino problem, The solar atmosphere, chromosphere and corona	2	
	10	The solar wind, The sun's magnetic field and the sunspot cycle, Prominences, flares and the interaction of the solar wind with the earth's atmosphere	3	
	11	The Hertzsprung–Russell diagram, The main sequence, The giant region, The white dwarf region, The stellar mass–luminosity relationship, Stellar lifetimes	3	
	Sections 2.2-2.8, 6.9, 6.12, 6.13 of Book 1			
III	STELLAR EVOLUTION		12	18

	12	Stellar Evolution, Low mass stars, Mid mass stars. Moving up the main sequence	2	
	13	The triple alpha process, The helium flash, Variable stars	3	
	14	Planetary nebula, White dwarfs, The discovery of white dwarfs, The future of white dwarfs, Black dwarfs, The evolution of a sun-like star, Evolution in close binary systems – the Algol paradox	3	
	15	High mass stars in the range >8 solar masses, Type II supernova, The Crab Nebula, Neutron stars and black holes,	2	
	16	The discovery of pulsars, What can pulsars tell us about the universe? Black holes, The detection of stellar mass black holes	2	
	Chapter 7 of Book 1			
IV	GALAXIES AND THE UNIVERSE		12	16
	17	The Milky Way, Open star clusters, Globular clusters, The interstellar medium and emission nebulae	2	
	18	Size, shape and structure of the Milky Way, A super-massive black hole at the heart of our galaxy	2	
	19	Other galaxies, Elliptical galaxies, Spiral galaxies, Evidence for an unseen component in spiral galaxies – dark matter, Irregular galaxies, The Hubble classification of galaxies	3	
	20	Active galaxies, Groups and clusters of galaxies, Superclusters, The structure of the universe	2	
	21	Big Bang models of the universe, The expansion of the universe, The cosmological redshift, The steady state model of the universe, Big Bang or Steady State?	2	
	22	The cosmic microwave background, The discovery of the cosmic microwave background, Inflation, Formation of the primeval elements	1	

	Sections 8.1-8.3, 9.2-9.9 of Book 1			
V	OPEN ENDED MODULE: MASTERING HASHING FOR EFFICIENT DATA HANDLING		12	
		• VO Tools. Reference 6 • Session 8.4, Question 1 of Book 1 • Planets, comets, asteroids etc. • Vizier, CDS, NED, SDSS etc. • Observing in other wavebands • Binary stars and Extra-solar Planets	12	
	References: Book 2-5			

Books and References:

1. Introduction to Astronomy and Cosmology by Ian Morison, John Wiley & Sons, 2008 (Book 1)
2. The physical universe: An introduction to astronomy, F.Shu, Mill Valley: University Science Books. (Book 2)
3. Modern Astrophysics, B.W. Carroll & D.A. Ostlie, Addison-Wesley Publishing Co.(Book 3)
4. Baidyanath Basu, 'An introduction to Astrophysics', Second printing, Prentice -Hall of India Private Limited, New Delhi, 2001.(Book 4)
5. Astronomy: A Physical Perspective by Marc L. Kutner, Cambridge University Press(Book 5)
6. <https://va-iitk.vlabs.ac.in/?page=listexp>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	2	3	1	1	2	3	2	3	1	2	3	3
CO 2	3	2	2	1	2	2	3	2	3	1	2	3	3
CO 3	2	2	2	2	2	2	3	2	3	1	2	3	3
CO 4	2	2	2	2	2	2	3	2	3	1	2	3	3
CO 5	3	2	2	1	2	2	3	2	3	1	2	3	3
CO 6	3	2	3	2	2	2	3	2	3	1	2	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	